The Bombay Salesian Society's

DON BOSCO INSTITUTE OF TECHNOLOGY

Department of Computer Engineering AY 26-27

25CE3CEP01 Community Engagement Project

Class:A	Batch:A	Group No.:3

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Experiment No. 3

Survey analysis and problem definitions

Aim: To determine survey methodology, conduct interviews, and provide analysis of results to make a feature list.

In Experiment No. 3, the problem will be defined after analyzing the survey/interview/research data.

CONDUCT INTERVIEW/ SURVEYS

Next, prepare questions for the interviews and/or surveys.

NOTE: Questions must be approved by guide before sharing the survey links or conducting interviews

Questionnaires must be designed section-wise.

1. Find out if your selected problem is something that people are facing or not. This speaks to the validity of the problem as a community problem.
2. Mention that you are developing a solution to the problem and list the features. Ask them if the features would help them solve the problem that they are facing.

3. Ask them if they will use your solution and ask for their suggestions regarding the platform, features, and functionality of the solution.

For interviews: (Max 3 people interviewed on video 5-10 mins each) - Minimum 10-15 questions
For surveys: Google forms and similar methods. Minimum 20 (total) questions divided in sections.

Provide justification for questions selected. Explain how they impact the research, understanding of the target audience, and how by conducting the survey you will understand how to design the solution.

Survey Question	**Why did we select it?**	**How does it impact our research/design?**
What work do you do?	Students, employees, commuters, etc. have different travel patterns and crowd exposure.	Helps identify our major **target audience** and the environments where the solution is most useful.
Have you ever felt uncomfortable, anxious, or claustrophobic while travelling in a crowded environment?	Directly measures the **human impact of overcrowding**.	If many users experience discomfort, it validates the need for real-time crowd monitoring and early warnings.
How frequently do you experience stamped/chaotic crowd situations?	Measures how often users encounter dangerous or uncontrolled crowd conditions.	Helps establish the **severity and frequency of the problem** and whether continuous monitoring is justified.
When a train or coach feels extremely crowded, what do you usually do?	Understands the current coping behaviour of users.	If people wait, change coaches, leave, or avoid travelling, the system could provide **early crowd information** so they can make better decisions before entering.
Have you ever avoided a particular coach, platform area, or train because of crowding?	Identifies whether overcrowding actually changes people's travel decisions.	Supports features such as **crowd-level displays, coach-wise crowd information, and route/coach recommendations**.
How important is privacy when using technology to monitor crowds?	Privacy is one of the major concerns with conventional camera-based monitoring.	Helps determine how strongly we need to communicate the **privacy advantage of Wi-Fi CSI**, which does not require identifying individual faces.

Survey Question	Why did we select it?	How does it impact our research/design?
What would make you trust a non-privacy-invasive crowd monitoring system?	Simply knowing that a system is "privacy-friendly" may not be enough for user acceptance.	Helps identify trust factors such as **no cameras, no personal identification, transparency, data security, or limited data storage**. These can become design requirements.
How frequently do structural obstructions such as pillars, walls, etc. affect your movement/experience?	Crowded spaces often contain physical obstacles that affect movement and sensing.	Helps us study whether our system needs to account for **complex layouts and obstructed areas**, rather than assuming an open environment.
Would you prefer a crowd-monitoring system that requires no smartphone/device?	This directly tests acceptance of **device-free sensing**, which is a key feature of our Wi-Fi CSI approach.	If users prefer not to carry/install/use a device, it strongly supports our choice of **device-free crowd detection**.
Do you think event-planning/disaster management will be easier if there is real-time crowd information?	Tests the usefulness of crowd monitoring beyond normal commuting.	Helps us identify **secondary users** such as event organizers, railway/metro authorities and disaster-management teams.
Do you feel knowing beforehand the busy/crowd status of public places would be useful?	Measures demand for **predictive/advance crowd information** rather than simply detecting crowds after they occur.	Can justify features such as **pre-entry crowd warnings, crowd forecasts and live status displays**.
How does "Device-free crowd monitoring" sound to you?	Measures the initial perception and acceptance of our proposed concept.	Helps understand whether the concept is easily understandable and attractive to the target audience.
In metros/locals, an app/display showed Low, Medium, High crowd levels — would you find this useful?	Tests a **specific proposed feature**, rather than asking only general opinions.	Directly helps us decide whether to implement a simple **Low → Medium → High crowd indicator**, which can be displayed on an app/dashboard/station screen.

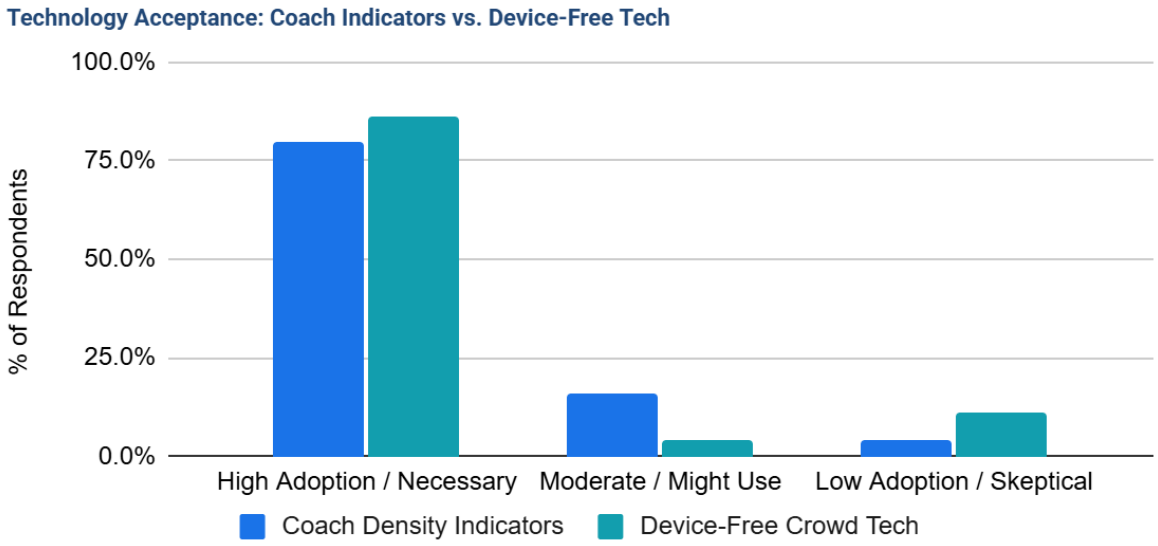
ANALYSIS REPORT

Include an **analysis report** of the surveys/interviews after you get your responses. 150-200 is a good number of responses.

TRAIN AVOIDANCE	PRIVACY SENSITIVITY	DEVICE-FREE PREFERENCE
62.0%	96.0%	80.0%
Avoided coach / platform	*Demand privacy protection*	*Favor non-phone / GPS tracking*

TECHNOLOGY VIABILITY & PRIVACY GUARDRAILS
• Overwhelming Demand for Indicators: 80.0% would actively use Low / Med / High coach displays.
• Strong Device-Free Mandate: 80.0% prefer crowd sensing without smartphone connection or GPS tracking.
• High Privacy Sensitivity: 96.0% prioritize privacy; 76.0% require secure data handling & 75.0% clear policy.
• High Utility Recognition: 86.0% affirm device-free technology is necessary for public safety & crowd alerts.
• Strategic Recommendation: Deploy anonymous, device-free density displays to maximize public trust.

COMMUTER PAIN POINTS & TRANSIT FRICTION
• Acute Overcrowding Stress: 88.0% of commuters experience severe discomfort, anxiety, or claustrophobia.
• Significant Avoidance Behavior: 62.0% regularly bypass crowded coaches or trains, sacrificing efficiency.
• Passive Coping vs. Delay: 67.0% cope by waiting for the next train; 25.0% continue travelling in distress.
• Real-Time Info Demand: 77.0% value knowing station and platform crowding levels prior to arrival.
• Demographic Core: 87.0% of respondents are Students (44%) & Working Professionals (43%) aged 19-50.



FEATURE LIST

Based on your observations from analyzing the survey responses, conclude your report with a feature list that will be implemented in your solution.
1.Emergency alerts/ crowd navigation
2.Admin portals (For event managers/disaster team)
3.Crowd density indicator

PROBLEM STATEMENT

Finally, **draft a Problem Statement.** Problem Statement is framed in the following manner:

To develop a website that provides device-free, privacy-preserving crowd management and navigation.